

Geometry of Splats

Rotskoff et al. 2019: *conditional* global optimization of least squares risk over mean-field two layer net via W+FR dynamics.

Quantitative $O(t^{-1})$ decay of optimality gap, due to FR!

Strong evidence for global optimization of splat models!

Parameter Estimation

Preliminary minimax analysis via reduction to *Gaussian location mixtures*.

$$\mathcal{A}_k := \left\{ \frac{1}{k} \sum_{i=1}^k \delta_{b_i} : b_1, \dots, b_k \in \mathbb{R} \right\} \quad \bar{\mathcal{G}}_k := \{s * \rho : s \in \mathcal{A}_k\}$$
$$\rho = \mathcal{N}(0,1)$$